

AUTOBIOGRAPHICAL REFLECTION • REQUIREMENT (III)

FROM ROOT TO ALGORITHM

Identity, Software Engineering, and Systems Innovation in East Africa

Simon Wamugunda Waweru

| BBIT Candidate • Dedan Kimathi University of Technology

Foundations in Kiambu & Nyeri

Operational Ecosystems

Early exposure to agrarian supply chains in Nyeri and fast-paced commercial trade in Wangige shaped a deep systemic awareness.

Efficiency & Order

Observed how manual bookkeeping bottlenecked informal trade, laying the mental groundwork for modern enterprise SaaS design.

CORE VALUES

The Family Problem-Solving Ethos



Analytical Assessment

Deconstruct complex obstacles into logical components.



Root-Cause Fixing

Target underlying causes instead of temporary patches.



Resource Efficiency

Maximize impact using lean available inputs.

Mathematical Logic & Leadership

100%

Algorithmic Logic Focus

Applied abstract mathematical proofs directly to program design.

Leadership Development

Peer Mentorship: Organized academic study circles and problem-solving labs.

Conflict Resolution: Mediated student organization dynamics effectively.

Clear Communication: Aligned technical goals with group consensus.

Academia at Dedan Kimathi University (DeKUT)



Software Engineering

JavaScript, Python, React, and Node.js full-stack paradigms.

Relational Databases

ACID transactions, SQL query optimization, and schema design.

Agile SDLC Workflows

Git branching, peer code reviews, and rapid sprint cycles.

Data Engineering at Trash Handlers Ltd

Operational Cleaning

Standardized raw weighbridge logs across plastic, paper, and metal streams.

Executive Reporting

Automated weekly yield dashboards for real-time manager decisions.



Architectural Software Portfolio

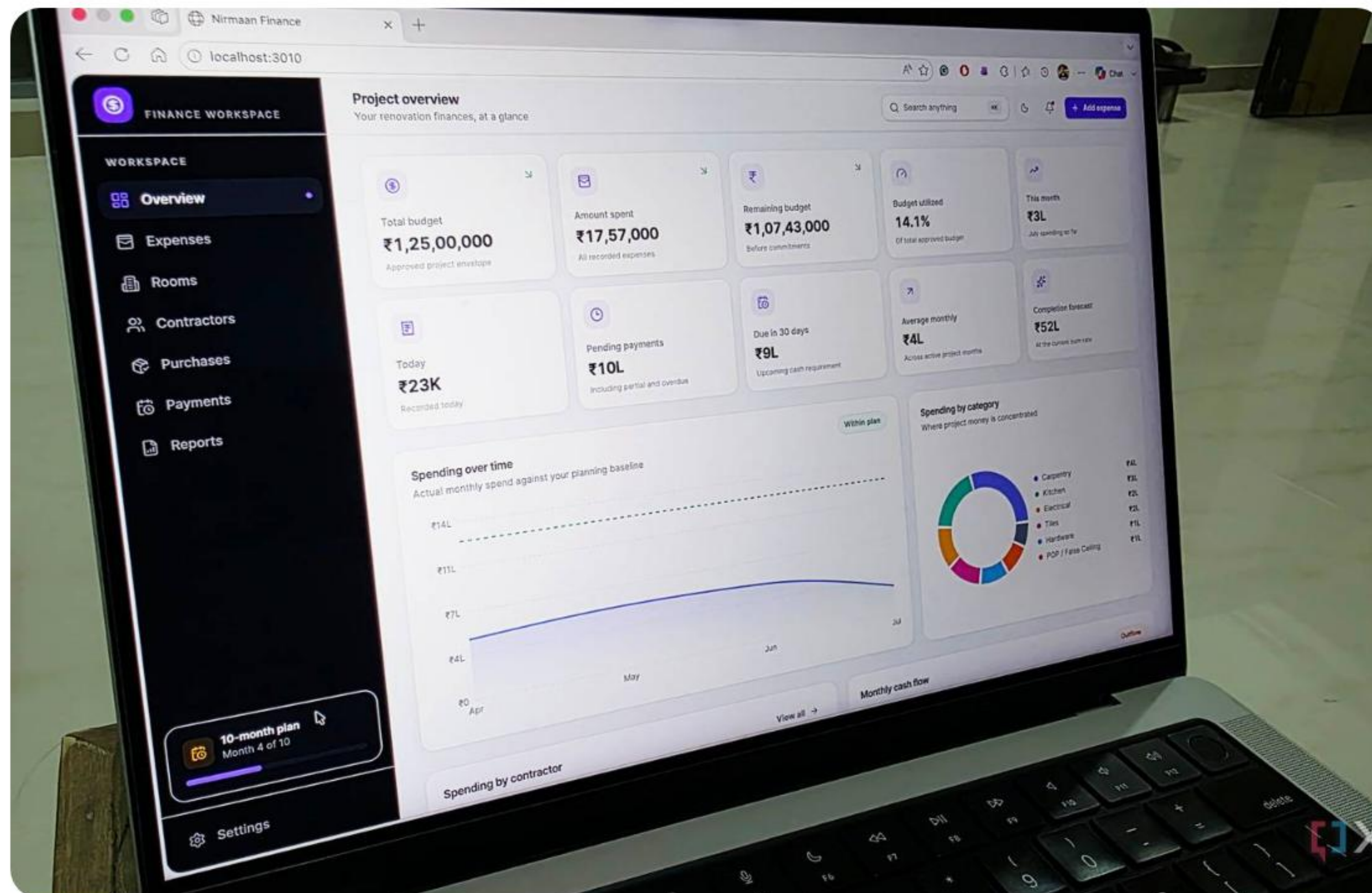
4+

Deployed Full-Stack & SaaS Systems

Architected for bandwidth efficiency, mobile responsiveness, and high data integrity across East African business environments.

CASE STUDY 01

GreenPlanet-MRF Platform



Intake Tracking

Real-time weight categorization across multi-bay facilities.

Stock Classification

Automated sorting yield logs linked to buyer invoicing.

Yield Analytics

Exportable executive reports eliminating manual data loss.

Instakill-Fumigation Platform



Lighthouse Speed

Optimized lightweight code delivering high performance on 3G networks.



Dynamic Booking

Interactive quote engine reducing client onboarding friction.

Ethics & Gender Equity in Tech



Ethical Data Privacy

Safeguarding user security and preventing predatory digital practices.



Inclusive UX Design

Designing interfaces tailored to female micro-entrepreneurs in Kenya.

10–Year Strategic Horizons

2026 – 2028

Technical Mastery

Complete BBIT degree and master cloud-native microservices architecture.

2028 – 2032

SaaS Enterprise

Launch East African SME software firm for logistics and recycling hubs.

2032 & Beyond

Ecosystem Legacy

Drive continental digital scale and launch youth tech mentorship hubs.

CONCLUSION & Q&A

CODE WITH PURPOSE & INTEGRITY

Building resilient software systems to power Africa's digital economy.



Simon Wamugunda Waweru

0757 153 838 | simnwamugunda25@gmail.com

Image Sources



<https://www.dkut.ac.ke/images/DeKUT-Nominated-as-Best-Performing-Institution-by-the-Public-Service-Commission-March-2025.png>

Source: www.dkut.ac.ke



<https://hnxzmedhuxtcuwoidkcr.supabase.co/storage/v1/object/public/news-media/articles/contamination-mrf-sorting-line.jpg>

Source: www.calwaste.com



<https://static0.xdaimages.com/wordpress/wp-content/uploads/wm/2026/07/codex-vs-claude-code-vs-github-copilot.jpg?w=1600&h=900&fit=crop>

Source: www.xda-developers.com